Unit 1 – MFT

1. Assume the R is a relation on a set A. aRb is partially ordered such that a and b are
   a) Reflexive
   b) Transitive
   c) Symmetric
   d) Reflexive and Transitive
2. A regular language over an alphabet Σ is one that cannot be obtained from the basic languages using the operation:
   a) Union
   b) Concatenation
   c) Kleene*
   d) All of the mentioned
3. The minimum number of states required to recognize an octal number divisible by 3 are/is:
   a) 3
   b) 4
   c) 5
   d) The number of elements in the set for the Language
4. L={x∈Σ* | "length if x is at most 2"} and Σ={0,1} is:
   a) ∅
   b) {ε}
   c) Finite
   d) Infinite
5. Given Σ= {a, b}, L= {x∈Σ*|x is a string combination}. Σ* represents which among the following?
   a) {aa, ab, ba, bb}

b) {aaa, aab, aba, abb}

c) {aaa, aab, aba, abb, bbb}

d) All of the mentioned

6. The Grammar can be defined as: G= (V, Σ, P, S). In the given definition, what does S represent?

   a) Accepting State

   b) Starting Variable

   c) Sensitive Grammar

   d) None of these

7. State true or false:

   Statement: R->RT

   T->a is an ambiguous grammar

   a) true

   b) can't say

   c) false

   d) none

8. What is the output for the given language?

   Language: A set of strings over Σ = {a, b} & takes as input and it prints 1 as an output "for every occurrence of a, b as its substring."

   (INPUT: abaabb)

   a) 01010010

   b) 01010100

   c) 01101010

   d) 01010101

9. A language for which a DFA exists is a:

   a) Regular Language

   b) Non-Regular Language

c) May be Regular

d) Cannot be said

10. When there are 2 finite states equivalent?

a) Same number of transitions

b) Same number of states

c) Same number of states as well as transitions

d) Both are final states

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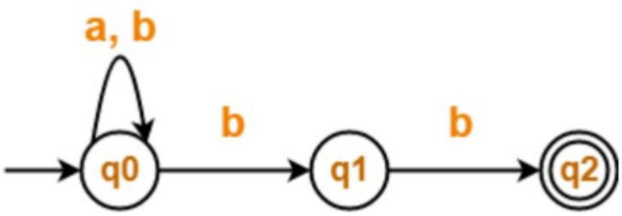
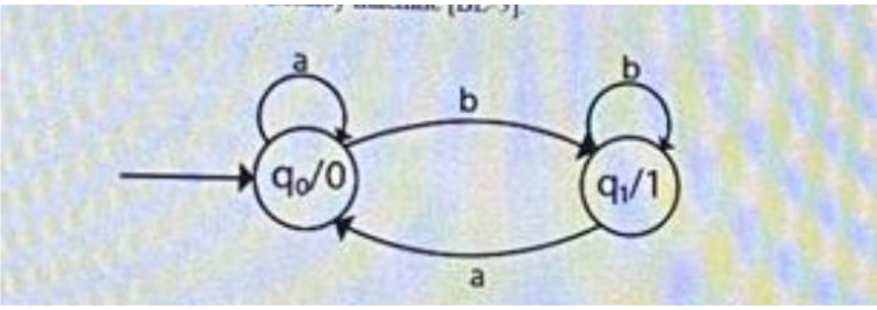
11. Explain transition diagram, transition table and transition function of DFA with suitable examples. [BL-2]

12. Define ε-transitions and convert the following NFA to DFA: [BL-3]

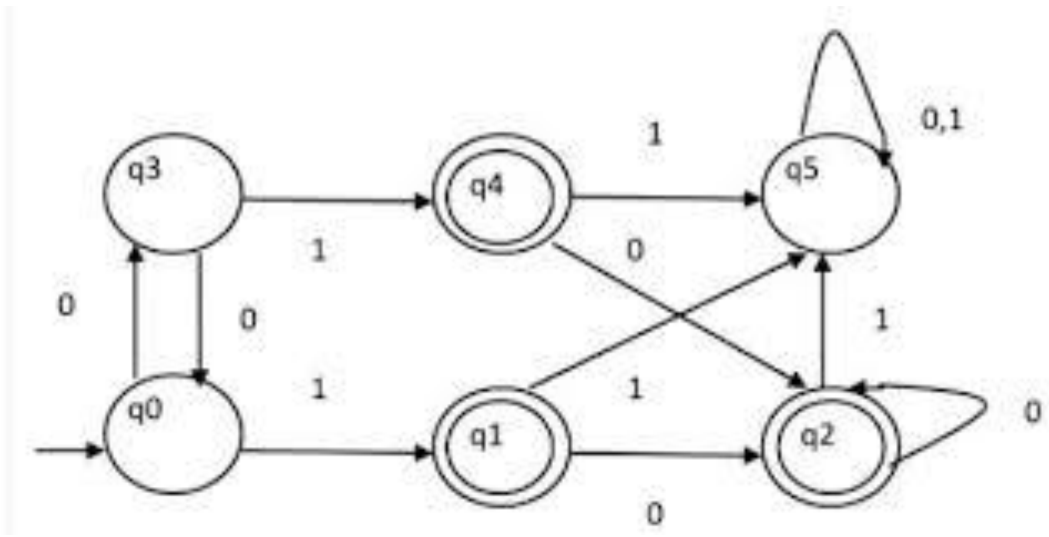


13. Define DFA, NFA, String, Language and Grammar with suitable explanation [BL-2]

14. Differentiate between Mealy and Moore machine also convert the following Moore machine into Mealy machine .[BL-3]



15.Minimize the below given DFA [BL-3]

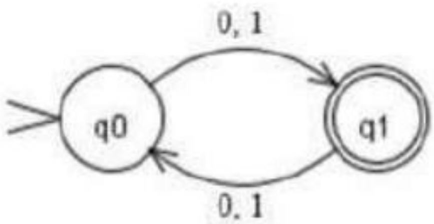


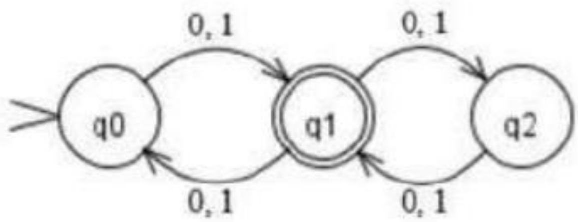
16. Obtain DFAs to accept strings of a's and b's having exactly one a. [BL-3]

17.Draw a DFA for a string ending in '01' and 1 ending with 011. [BL-3]

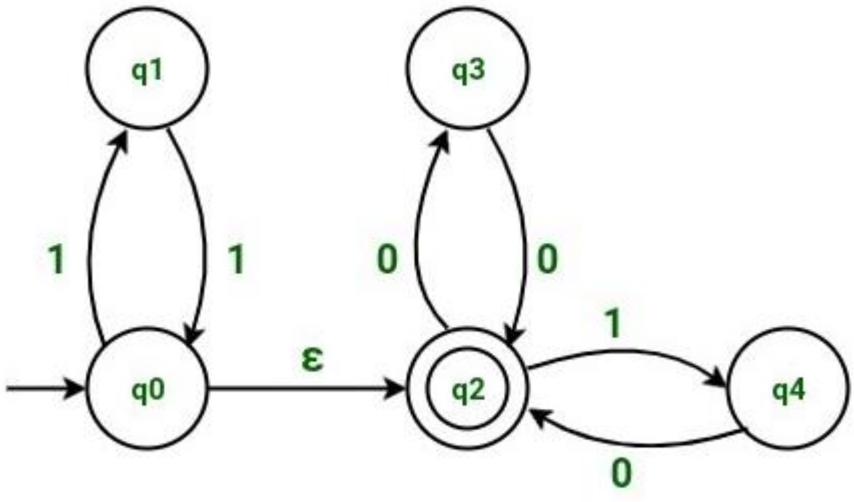
18.Design a finite automata which performs addition of two binary numbers also Draw a DFA to accept string of 0's and 1's ending with 011. [BL-3]

19.Proof that these two machines are equivalent or not? [BL-3]





20. Convert the following epsilon-NFA to NFA [BL-3]



21.Explain Myhill Nerode theorem with suitable example in detail [BL-3]